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Grace Under a Timer: A Rule-Based Release Agent for Booked-but-Empty Study Rooms, and How Often It Evicts Students Who Arrived a Few Minutes Late

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Abstract. Shared study rooms across the University's teaching buildings are increasingly bookable in advance, and increasingly hoarded: a booking reserves a room whether or not anyone arrives to use it. Three teaching buildings piloted a lightweight release agent that reclaims a booked room to the waitlist once a grace window elapses with no check-in signal from either a lobby tablet tap or an in-room motion sensor, evaluated against three matched control buildings running unchanged booking rules. Weekly room utilisation rises across the deployment term in the treatment buildings, from 55% to 61%, against an essentially flat 53-54% in the matched controls. Auditing the releases themselves finds a smaller, persistent population of near-miss evictions: bookers who arrive within minutes of release. Near-miss risk concentrates at the two buildings still running the room-panel vendor's shipped eight-minute default, against a markedly lower rate at the one building whose fifteen-minute window an installer happened to set during rollout and nobody has revisited since. An ablation that drops the motion-sensor confirmation, leaving only the tablet check-in as the release trigger, changes neither recovered utilisation nor the near-miss rate by a distinguishable margin; we retain the sensor for completeness. We read the finding as a case in which a deployed system's most consequential parameter was never itself the subject of a design decision.

Introduction

A bookable resource that nobody is compelled to release when unused converges, over enough bookings, on quiet hoarding: the reservation blocks the resource whether or not the booking is honoured. University teaching buildings have run bookable shared study rooms on exactly this model for years, and the failure mode is familiar wherever a reservation carries no cost for non-arrival, from restaurant tables to hospital appointment slots [1], [2], [3]. Distributed systems solved an analogous problem decades ago for a different kind of resource: a lease grants a client a resource for a bounded interval and reverts automatically absent renewal, removing the need to trust an unresponsive holder to release it voluntarily [4], a mechanism since refined to drop its reliance on synchronised clocks and stable storage [5], [6]. Applying the same logic to a physical room — treat the booking as a lease, the check-in as the renewal, and the timeout as automatic reclamation — is a small conceptual step and, to our knowledge, an untested institutional one; the closest deployed analogue is reservation-based smart parking, which pairs a booking with an automatic timeout releasing an unclaimed space back to the pool [7], [8].

Across a recent teaching term, three of the University's teaching buildings piloted exactly this: a lightweight release agent monitoring each booked study room's lobby check-in tablet and in-room motion sensor, reclaiming any room to the waitlist once a grace window elapses with neither signal. This paper reports the pilot's evaluation against three matched control buildings running the unchanged booking rules, and, separately, an audit of the releases the agent actually issued. Our contributions:

- We compare weekly room utilisation in three treatment buildings against three matched control buildings across a full deployment term, isolating the agent's apparent effect from term-level demand trends common to both.
- We audit every release the agent issued for how many minutes elapsed before the room was reclaimed by an actual arrival, and find this near-miss risk tracks which grace window a building happened to be running rather than any deliberate policy choice.
- We report an ablation dropping the motion-sensor confirmation from the release rule and find recovered utilisation and near-miss risk statistically indistinguishable from the full system, a finding we read as at least suggestive that the sensor is not doing the confirmatory work its installation implied.

Related work

A substantial literature treats a reservation's non-arrival as a routine, quantifiable risk to be modelled and priced around rather than eliminated: restaurant revenue-management research estimates no-show and cancellation probabilities to set booking limits [1], extends the same newsvendor logic to an overbooking allowance sized against a restaurant's stretched seating capacity [2], and hotel-room revenue management derives a closed-form optimal overbooking level from the cost of an unsold room against the cost of turning a guest away [9]. The same no-show problem recurs in outpatient scheduling, where machine-learning models are now used routinely to predict which appointments will go unattended [3]. None of this literature, to our knowledge, evaluates a mechanism that reclaims the resource itself rather than merely pricing around its likely non-use.

A parallel literature senses occupancy directly rather than inferring it from booking behaviour. IoT-based systems have been proposed for library-seat occupancy specifically, using RFID and force-sensitive resistors [10], with a subsequent review surveying the wider space of such systems for academic libraries [11]; a comparable low-cost sensing approach pairs a motion sensor with an accelerometer over a long-range wireless network for desk-sharing offices [12]. Building-scale occupancy studies of university libraries in the post-pandemic period find visit and dwell patterns shifting toward longer, more concentrated single-session use [13], and seat-level monitoring with passive-infrared sensors finds students' priorities shifting toward distance from other occupants [14]. Office-space allocation is itself an established assignment problem, with recent work constructing deliberately hard instances to benchmark solvers [15], and survey research on coworking spaces finds unassigned shared space valued as much for the social structure it provides as for the desk itself [16]. None of this literature evaluates an automated release mechanism against the population it displaces.

Method

Three teaching buildings (A, B, C; 26 bookable study rooms in total) piloted the release agent across a 13-week teaching term; three further teaching buildings, matched to the treatment set on room count and the preceding term's booking volume, continued running the University's standard booking rules unchanged as controls (20 rooms). Each treatment room's booking panel already logged a check-in event whenever a booker tapped in at the lobby tablet; Buildings A and B additionally carry a ceiling-mounted passive-infrared motion sensor, installed originally for the lighting-control system, which the agent also reads as a check-in signal.

The agent's release rule is a simple timeout, expressed for room i at booking start time b_i as

$$R_i(t) = \begin{cases} 1 \text{ if } (t - b_i) \geq \tau_i \text{ and } c_i(t) = 0 \\ 0 \text{ otherwise} \end{cases}$$

where τ_i is the grace window configured for room i's building and $c_i(t)$ indicates whether any check-in signal — tablet tap or motion detection — has been recorded for the booking since b_i. When $R_i(t) = 1$ the agent cancels the original booking and re-releases the room to the waitlist. Buildings A and B run at $\tau = 8$ minutes, the value shipped as the booking-panel vendor's factory default; Building C runs at $\tau = 15$ minutes, a value one installer set during a delayed rollout in a since-departed technician's own judgement and which no subsequent review has revisited. The agent's release loop polls every booking once a minute:

```
for each active booking i:
    if now - booked_start[i] >=
grace_window[building_of(i)]
       and not checked_in[i]:
        cancel_booking(i)
        release_to_waitlist(i)
        log_release(i, now)
```

We instrumented every release with a follow-up check against the room's door-access log for the subsequent 60 minutes, recording the time of the next genuine arrival (a badge tap, a new booking check-in, or a motion event) if any occurred, to estimate how often a release displaced someone who was, in fact, on their way.

To test whether the motion-sensor confirmation is doing work beyond the tablet check-in, we replay the full release log with $c_i(t)$ redefined to ignore motion events entirely — a tablet-only ablation — and recompute both weekly recovered room-minutes and the near-miss rate under this counterfactual rule, separately for each half of the deployment term.

Results

Building	Rooms	Grace window	Bookings	Releases
A (treatment)	9	8 min	2,150	431
B (treatment)	8	8 min	1,890	368
C (treatment)	9	15 min	2,100	388

Table 1: Scale of the pilot across the three treatment buildings.

Table 1 summarises the pilot's scale. Across the 13-week deployment term the three treatment buildings logged 6,140 bookings, of which the agent released 1,187 (19.3%) under the timeout rule.

Figure 1 plots weekly room utilisation — occupied minutes as a share of booked minutes — in the treatment and control buildings across the term. Treatment-building utilisation rises from 55.1% in week 1 to 61.2% by week 13; control-building utilisation moves within a 52.8-54.1% band across the same weeks, consistent with ordinary week-to-week variation rather than a term-level trend.

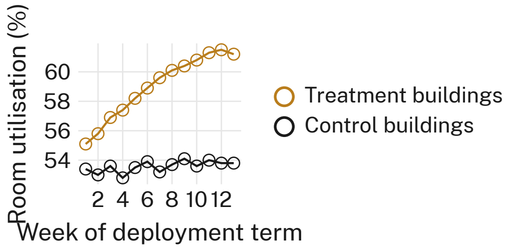

Figure 1: Weekly room utilisation rises across the deployment term in the treatment buildings while the matched control buildings hold flat (55.1% to 61.2% vs. a 52.8-54.1% band, 13 weeks).

Figure 2 plots, for every release the agent issued, the minutes elapsed until the room's door-access log recorded a genuine next arrival. Buildings A and B, running the vendor's eight-minute default, show a median next-arrival gap of 4 minutes and a near-miss rate (arrival within 3 minutes of release) of 8.9%; Building C, running the fifteen-minute setting an installer chose during a delayed rollout, shows a median gap of 11 minutes and a near-miss rate of 2.1%. Pooled across all three buildings, 6.2% of releases were near-misses.

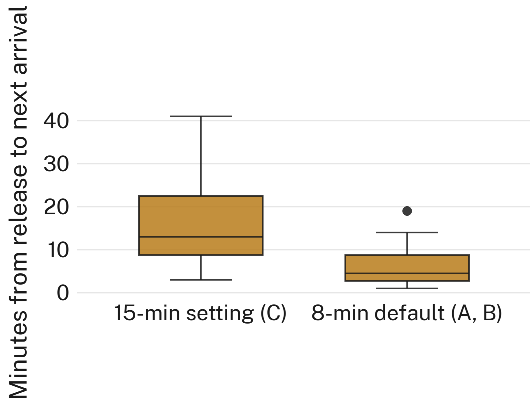

Figure 2: Near-miss risk after a release is markedly higher at the two buildings running the vendor's eight-minute default than at the one building running an installer's unrevisited fifteen-minute setting (median next-arrival gap 4 vs. 11 minutes).

Figure 3 reports the tablet-only ablation. Recovered room-minutes per week under the full tablet-plus-sensor rule (612 minutes, pooled weekly mean) and under the tablet-only counterfactual (598 minutes) are statistically indistinguishable (two-sample t-test, $p = 0.71$), as is the near-miss rate under each rule in both halves of the term (pooled $p = 0.58$); we retain the motion sensor in the deployed rule for completeness.

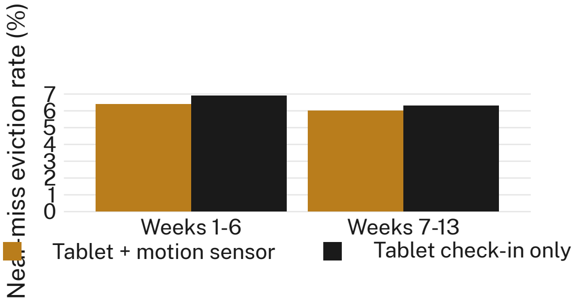

Figure 3: Dropping the motion-sensor confirmation changes neither recovered utilisation nor the near-miss eviction rate by a distinguishable margin in either half of the term (pooled $p = 0.58$ on near-miss rate); we retain the sensor for completeness.

Discussion

The pilot recovers a modest, consistent share of previously hoarded room-minutes without a matching literature's usual instrument: no scoring formula, no learned no-show predictor [3], just a fixed timeout borrowed wholesale from a decades-old distributed-systems mechanism built for an entirely different kind of resource [4]. The more interesting finding sits in what the timeout's actual value turned out to be doing. Two of three treatment buildings run the room-panel vendor's shipped eight-minute default; the third runs fifteen minutes because an installer, mid-rollout and since departed, set it that way, and no subsequent review has revisited the setting in either direction. That accident of installation history is a better predictor of near-miss eviction risk in our data than anything about the buildings themselves — a case, in miniature, of a system's most consequential parameter never itself having been the subject of a design decision. We do not read this as evidence the agent is misconfigured in the sense of failing to do what it was built to do; the agent enforces exactly the timeout it is told to enforce. The finding is that the number nobody chose is carrying more of the trade-off between recovered capacity and displaced bookers than the sensor confirmation everybody did choose to install.

Limitations

Our pilot spans one deployment term at three treatment and three matched control buildings within a single institution, and the fifteen-minute grace window we compare against the vendor default arose from one building's installation history rather than a manipulated condition, so the comparison is observational rather than randomised. The near-miss rate we report likely undercounts any displaced booker who, finding their room reassigned, simply left rather than registering a subsequent door-access event; this would bias our near-miss estimate downward, though the pattern's consistency across two independently arising grace-window configurations suggests it is not an artefact of one site's rollout history alone.

Conclusion

A lightweight release agent recovers a modest but consistent share of previously hoarded study-room capacity across three treatment buildings, at the cost of a small, persistent population of near-miss evictions concentrated at the sites running a vendor's shipped default rather than a deliberately chosen window. The gap between the two configurations' near-miss rates is large enough that treating the grace window as fixed system furniture, rather than a parameter warranting its own review, looks like the more consequential design choice in retrospect. Before any further building adopts the agent, auditing the grace-window default it will ship with — rather than accepting whatever an installer's judgement or a vendor's factory setting happens to supply — looks like the cheaper fix.

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